STACKS, DISCRIMINANTS: LEMMA-DUALIZING-WELL-DEFINED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 356-360 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 362-377, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 33 of queue_discriminant.json (pb=694, stmt 356-360, proof 362-377).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BSZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 308–310. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT0: sources/stacks-discriminant/pinned-original-source.tex, source lines 321–329. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The module (0BSZ) is well defined, i.e., independent of the choice of the factorization.*

Proof. Let B', B'', B''' be as in Lemma 0BT0. We obtain a canonical map

$$\omega''' = \mathrm{Hom}_A(B''', A) \otimes_{B'''} B \longrightarrow \mathrm{Hom}_A(B', A) \otimes_{B'} B = \omega'$$

and a similar one involving B''. If we show these maps are isomorphisms then the lemma is proved. Let $g \in B'$ be an element such that $B'_g \to B_g$ is an isomorphism and hence $B'_g \to (B''')_g \to B_g$ are isomorphisms. It suffices to show that $(\omega''')_g \to \omega'_g$ is an isomorphism. The kernel and cokernel of the ring map $B' \to B'''$ are finite A-modules and g-power torsion. Hence they are annihilated by a power of g. This easily implies the result. □

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.